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Flip-chip light emitting diode and manufacturing method thereof

Abstract

A flip-chip light emitting diode includes a light-emitting epitaxial laminated layer having a first surface, and a second surface opposing the first surface, and including a first semiconductor layer, an active layer, and a second semiconductor layer, wherein the first surface is a roughened surface; a transparent bonding medium layer over the first surface of the light-emitting epitaxial laminated layer and bonded with a transparent substrate; and a first electrode and a second electrode respectively over a first electrode region and a second electrode region over the second surface of the light-emitting epitaxial laminated layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of U.S. patent application Ser. No. 17/337,361 filed on Jun. 2, 2021, which is a continuation of U.S. patent application Ser. No. 16/221,706 filed on Dec. 17, 2018, which in turn is a continuation of, and

claims priority to, International Application No. PCT/CN2017/085661 filed on May 24, 2017, which claims priority to Chinese Patent Application No. 201610450480.0 filed on Jun. 22, 2016. The disclosures of the above-referenced applications are hereby incorporated by reference in their entirety.

BACKGROUND

(1) In recent years, light emitting diodes (LEDs) are widely applied and plays an increasingly important role in various applications such as a display system, a lighting system and an automobile tail light.

(2) With the successful development of epitaxial growth technique of organometallic chemistry, AlGaInP-based materials are developed and used for manufacturing high-power and high-luminance LEDs that emits red light and yellow light.

SUMMARY

(3) The inventors of the present application have recognized that, a thick substrate is prone to absorbing light from the light source, and a thin chip may result in chip crack and poor yield. In a general process structure, the transparent substrate is required to be $>80\text{ }\mu\text{m}$, and the surface is a polishing plane. Therefore, in the flip-chip quaternary light emitting diode structure as shown in FIG. 1, light from the radiation light source is readily absorbed by material in the substrate, and the light emitting surface is likely to reflect light back to the active layer structure, causing light loss.

(4) In an aspect, a method is provided for manufacturing a quaternary wafer chip having a patterned transparent substrate on the surface, which solves the above problems and enhances operation feasibility and light emitting efficiency.

(5) In some embodiments, the manufacturing method for a flip-chip light emitting diode includes: (1) providing a transparent substrate and a temporary substrate, and bonding the transparent substrate with the temporary substrate; (2) grinding and thinning the transparent substrate; (3) providing a light-emitting epitaxial laminated layer, having a first surface and a second surface opposite to each other, which further comprises a first semiconductor layer, an active layer and a second semiconductor layer; (4) forming a transparent bonding medium layer over the first surface of the light-emitting epitaxial laminated layer, and bonding the transparent bonding medium layer with the transparent substrate; (5) defining a first electrode region and a second electrode region over the second surface of the light-emitting epitaxial laminated layer, and manufacturing a first electrode and a second electrode; and (6) removing the temporary substrate.

(6) In some embodiments, step (1) further includes: patterning the first main surface of the transparent substrate and bonding the first main surface, which serves as the bonding surface, with the temporary substrate.

(7) In some embodiments, the transparent substrate should be able to be bonded with the temporary substrate via adhesive with high temperature resistance and good stability in step (1), and the temporary substrate can be made of glass, ceramic or sapphire.

(8) In some embodiments, the thinned transparent substrate in the step (2) should be thinner than $80\text{ }\mu\text{m}$.

(9) In some embodiments, the transparent bonding medium layer formed in the step (4) is a combination of one or more layers of silicon oxide layer, silicon nitride layer and aluminum oxide layer, such as: $\text{Al.sub.2O.sub.3/SiO.sub.2}$, $\text{SiN.sub.X/SiO.sub.2}$, $\text{SiN.sub.X/Al.sub.2O.sub.3/SiO.sub.2}$.

(10) In some embodiments, step (3) also includes the patterning of the first surface of the light-emitting epitaxial laminated layer. Step (4) also includes: after forming a transparent bonding medium layer over the first surface of the light-emitting epitaxial laminated layer, flattening the surface of the transparent bonding medium layer via chemical mechanical polishing, wherein, the roughness value $R_a < 1\text{ nm}$.

(11) In some embodiments, step (4) includes: forming a transparent medium layer over the first

surface of the light-emitting epitaxial laminated layer, wherein, the upper surface has grid-shaped or array-shaped recess region; depositing a patterned transparent bonding medium layer over the transparent medium layer, which fills up the recess region of the transparent medium layer, and the upper surface is at the same plane with the upper surface of the transparent medium layer.

(12) Further, in step 4), deposit a transparent bonding medium layer over the transparent medium layer and fine polish the surface till the transparent medium layer is exposed, hence the transparent bonding medium layer fills up the recess region of the transparent medium layer, and the upper surfaces of these two layers are at a same plane, and then polish the surfaces.

(13) In some embodiments, Step 5) also includes: arranging pyramid-shaped recess arrays over the second surface of the exposed light-emitting epitaxial laminated layer, wherein, each pyramid-shaped recess is arranged in the vertical shadow area of the transparent bonding medium layer to reflect light from the vertical shadow area of the transparent bonding medium layer; as a result, light can deviate from this shadow area, thus eliminating light reflection from the transparent bonding medium layer to the inside part of the chip.

(14) In another aspect, a flip-chip LED chip structure is provided, which can be manufactured by any of the abovementioned methods.

(15) Embodiments of the present disclosure can have one or more of the following advantageous effects: (I) the patterned substrate surface can change the angle of incidence of the totally-reflected light, which prevents light from being reflected back and re-absorbed by the semiconductor structure, thus increasing light extraction efficiency; (II) the thickness of the thin transparent substrate can be $<80\text{ }\mu\text{m}$, making operation yield more stable and avoiding light absorption by materials.

(16) The other features and advantages of some embodiments of the present disclosure will be described in detail in the following specification, and it is believed that such features and advantages will become more obvious in the specification or through implementations of this disclosure. The purposes and other advantages of the present disclosure can be realized and obtained in the structures specifically described in the specifications, claims and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To more clearly illustrate some of the embodiments of the disclosure, the following is a brief description of the drawings. The drawings are only illustrative of some embodiments, and for those of ordinary skill in the art, other drawings of other embodiments can become apparent based on these drawings.

(2) FIG. 1 illustrates a flip-chip light emitting diode chip according to some implementations.

(3) FIG. 2 is a flow diagram illustrating the manufacturing of a flip-chip light emitting diode according to some embodiments of the present disclosure.

(4) FIG. 3 is a schematic diagram illustrating a first step of the manufacturing of a flip-chip light emitting diode according to Embodiment 1.

(5) FIG. 4 is a schematic diagram illustrating a second step of the manufacturing of a flip-chip light emitting diode according to Embodiment 1.

(6) FIG. 5 is a schematic diagram illustrating a third step of the manufacturing of a flip-chip light emitting diode according to Embodiment 1.

(7) FIG. 6 is a schematic diagram illustrating a fourth step of the manufacturing of a flip-chip light emitting diode according to Embodiment 1.

(8) FIG. 7 is a schematic diagram illustrating a fifth step of the manufacturing of a flip-chip light emitting diode according to Embodiment 1.

(9) FIG. 8 is a schematic diagram illustrating a sixth step of the manufacturing of a flip-chip light

emitting diode according to Embodiment 1.

(10) FIG. 9 is a schematic diagram illustrating a seventh step of the manufacturing of a flip-chip light emitting diode according to Embodiment 1.

(11) FIG. 10 is a schematic diagram illustrating an eighth step of the manufacturing of a flip-chip light emitting diode according to Embodiment 1.

(12) FIG. 11 is a light path analysis diagram of the flip-chip light emitting diode of Embodiment 1 which improves light extraction efficiency.

(13) FIG. 12 shows the flip-chip light emitting diode chip of Embodiment 2.

(14) FIG. 13 shows the flip-chip light emitting diode chip of Embodiment 3, having a patterned transparent bonding layer.

(15) FIG. 14 is a top view of the patterned transparent bonding layer of the flip-chip light emitting diode as shown in FIG. 13.

(16) FIG. 15 illustrates the flip-chip light emitting diode of Embodiment 4.

IN THE DRAWINGS

(17) **110, 210**: transparent substrate; **120, 220**: transparent bonding layer; **130, 230**: p-type window layer; **140, 240**: P-type semiconductor layer; **150, 250**: active layer; **160, 260**: n-type semiconductor layer; **170, 270**: p electrode; **180, 280**: n electrode; **200**: growth substrate; **300**: temporary substrate; **310**: adhesive layer.

DETAILED DESCRIPTION

(18) The embodiments of the present disclosure will be described in detail with reference to the accompanying drawings and examples, to help understand and practice the disclosed embodiments, regarding how to solve technical problems using technical approaches for achieving the technical effects. It should be noted that the embodiments and their characteristics described in this disclosure may be combined with each other and such technical proposals are deemed to be within the scope of this disclosure without departing from the spirit of this disclosure.

(19) Transparent bonding can be realized for the quaternary LED by bonding a semiconductor structure with a transparent substrate, which can be manufactured in a horizontal structure, and then be extended to a flip-chip structure that matches with RGB applications, such a structure is illustrated in FIG. 1.

(20) In some embodiments, the first semiconductor layer and the second semiconductor layer are semiconductor materials of opposite types. For example, if the first semiconductor layer is an n-type semiconductor, then the second semiconductor layer can be a p-type semiconductor; and if the first semiconductor layer is a p-type semiconductor, then the second semiconductor layer should be a n-type semiconductor.

(21) The following embodiments disclose a method for manufacturing a quaternary wafer chip having a patterned transparent substrate on the surface. At first, bond the transparent substrate with a temporary substrate, and then conduct thinning process. Next, bond the transparent substrate with the epitaxy wafer via transparent bonding technology. Remove the growth substrate and manufacture the electrode. Finally, remove the temporary substrate to obtain a thin flip-chip light emitting diode, wherein, the transparent substrate plate can be as thin as less than 80 μm , making operation yield more stable and avoiding light absorption by materials.

(22) FIG. 2 illustrates a flow chart for manufacturing a flip-chip light emitting diode according to the present disclosure, which mainly comprises steps **S100-S600**. Details are illustrated below in combination with FIGS. 3-10.

(23) At first, provide a transparent substrate **210** and a temporary substrate **300**, and pattern the first main surface of the transparent substrate **210**. Bond the transparent substrate with the temporary substrate **300** via the adhesive layer **310**. The bonding surface of the transparent substrate is a patterned surface, which can be made of sapphire, aluminum nitride or glass. The patterned transparent substrate can be manufactured by different methods such as patterned sapphire substrate (PSS) technology; the temporary substrate **300** can be made of glass or ceramic; the

adhesive layer **310** can be made of adhesive with high temperature resistance and good stability, such as BCB adhesive, as shown in FIGS. **3** and **4**.

(24) Next, grind and thin the patterned transparent substrate **210** till less than 80 μm . The thinned transparent substrate is not readily broken thanks to the support from the temporary substrate **300**, which guarantees subsequent process. As shown in FIG. **5**, preferred thickness of the thinned transparent substrate **200** is 40-60 μm . In this embodiment, the thickness is 50 μm .

(25) Next, use a quaternary epitaxial wafer of MOCVD nucleation, the structure of which is as shown in FIG. **6**. A common structure is acceptable, which generally comprises a growth substrate **200**, an n-type semiconductor layer **260**, an active layer **250**, a p-type semiconductor layer **240** and a p-type window layer **230**.

(26) Next, deposit a transparent bonding layer **220** over the p-type window layer **230** of the epitaxial wafer, which is bonded with the thinned transparent substrate **210**, as shown in FIG. **7**. The transparent bonding layer **220** is a combination of one or more layers of silicon oxide layer, silicon nitride layer and aluminum oxide layer, such as: $\text{Al.sub.2O.sub.3/SiO.sub.2}$, $\text{SiN.sub.X/SiO.sub.2}$, $\text{SiN.sub.X/Al.sub.2O.sub.3/SiO.sub.2}$. For example, the transparent bonding layer **220** includes a first sub-layer with a relatively high refractive index formed on the p-type window layer **230**, and a second sub-layer with a relatively low refractive index formed on the first sub-layer. The first sub-layer has a thickness of t less than 1 μm , preferably less than 500 nm, and most preferably 100 to 300 nm; the second sub-layer has a thickness of more than 1 μm for example, preferably 1~3 μm . The second sub-layer is mainly used for bonding and connecting the transparent substrate **210** and the epitaxial layers, and can be one layer of silicon oxide layer. The first sub-layer can reduce the refractive index difference between the p-type window layer and the first sub-layer, thereby improving the light extraction efficiency. In this embodiment, $\text{Al.sub.2O.sub.3/SiO.sub.2}$ is used, and bonding is undertaken under high-temperature and high-pressure, wherein, high temperature is 360° C., and high pressure is 150 MPa.

(27) Next, remove the growth substrate **200** of the epitaxial wafer, and expose the surface of the n-type semiconductor layer **260**, as shown in FIG. **8**. In this embodiment, remove the growth substrate via wet etching, wherein, the substrate removal agent is prepared by NH.sub.4OH and H.sub.2O.sub.2 .

(28) Next, manufacture the p electrode **270** and the n electrode **280**, as shown in FIG. **9**. Specifically: define a p electrode region and an n electrode region over the n-type semiconductor layer surface. Etch the n-type semiconductor layer **260** and the active layer **250** of the p electrode region, and expose the p-type semiconductor layer **240**. Manufacture the p electrode **270** made of beryllium gold over the exposed surface, which is taken as the chip anode. Manufacture the n electrode **280** in the n electrode region of the n-type semiconductor layer **260**, which is taken as the chip cathode and also the reflector. In this embodiment, the material is Ag.

(29) Next, remove the adhesive with appropriate adhesive removal agent under appropriate conditions. The temporary substrate is also removed to expose the patterned transparent substrate surface, as shown in FIG. **10**. After splitting, the material can be used in the package structure of the flip-chip without grinding operation.

(30) FIG. **11** is an optical path analysis diagram of the flip-chip light emitting diode, which improves light extraction efficiency. The patterned substrate surface can change the angle of incidence of the totally-reflected light, which prevents light from being reflected back and re-absorbed by the semiconductor structure, thus increasing light extraction efficiency; meanwhile, the thickness of the thin transparent substrate can be $<80 \mu\text{m}$, making the operation yield more stable and avoiding light absorption by materials.

(31) FIG. **12** shows the flip-chip light emitting diode according to the second embodiment of the present disclosure. The difference of this embodiment and Embodiment 1 is that: rough the surface of the p-type window layer of the epitaxial wafer, which is taken as the bonding surface **230a**. Deposit the transparent bonding layer **220** over the bonding surface and bond the transparent

bonding layer with the transparent substrate. Specifically, wet etch the surface of the p-type window layer **230**, which forms a roughened surface **230a**. Next, deposit the transparent bonding layer **220** over the roughened surface **230a**, and flatten the surface of the transparent bonding layer **220** till its roughness value $Ra < 1$ nm, thus obtaining a flattened LED; in which, the surface is polished to required flatness in the manner of chemical mechanical polishing, with ESR-320 polishing agent and SUBA600SiO.sub.2 polishing pad.

(32) FIG. **13** shows the flip-chip light emitting diode according to the third embodiment of the present disclosure. The difference of this embodiment and Embodiment 2 is that: the patterned transparent bonding medium layer **220** is composed of a high-refractivity transparent medium layer **221** and a patterned transparent bonding layer **222**. The high-refractivity transparent medium layer **221** is over the surface of the p-type window layer **230**, wherein, the upper surface is provided with a grid-shaped or array-shaped recess region; a patterned transparent bonding layer **222** fills up the recess region of the high-refractivity transparent medium layer **221**, and the upper surface is at the same plane with the upper surface of the transparent medium layer **221**. Specifically, deposit the high-refractivity transparent medium layer **221** over the surface of the p-type window layer **230**, and define a circular array over the upper surface of the transparent medium layer **221** via photoetching. Etch the area within the circle to form an array-shaped recess region, as shown in FIG. **14**. Deposit a bonding layer **222** with thickness of 1-2 μ m over the transparent medium layer **221**. Then, fine polish the surface of the bonding medium layer **222** till the transparent medium layer **221** is exposed. In this embodiment, the high-refractivity transparent medium layer **221** is made of Al.sub.2O.sub.3, with thickness of 3 μ m. In the circular array, circle diameter is 3-6 μ m, circle center distance is 5-10 μ m, and etching depth is 1-2 μ m. The bonding layer **222** is made of SiO.sub.2.

(33) In this embodiment, the light-emitting medium of the semiconductor surface is changed to high-refractivity transparent material, which greatly increases light extraction efficiency from the semiconductor to the transparent substrate. Light emitted to the high-refractivity transparent medium is divided into two portions. Some light, such as light **1**, is emitted to the bonding layer **222**, and reflection, refractivity or total reflection take place over the lower surface. This portion of light is similar to that of the original bonding structure; some light is directly emitted to the transparent substrate through the high-refractivity transparent medium, which increases light into the transparent substrate, and therefore increases probability of light extraction, such as light **2**. In this case, the transparent bonding layer **222** also serves as a reflector, which partially reflects light back to inside part of the chip from the upper surface of the transparent substrate to prevent it from being emitted to and absorbed by the semiconductor layer. To sum it up, this embodiment provides a flip-chip light emitting diode with a patterned transparent bonding layer, which effectively improves light extraction efficiency and guarantees bonding strength and yield.

(34) FIG. **15** shows the flip-chip light emitting diode according to the fourth embodiment of the present disclosure, and the difference between this embodiment and Embodiment 3 is that: the lower surface of the n-type semiconductor layer **260** is etched to form pyramid-shaped recess arrays **261**. Each pyramid-shaped recess corresponds to the vertical shadow area of the patterned bonding layer **222**, and the diagonal length of the bottom surface is less than or equal to the diameter of the circular recess bottom layer of the patterned bonding layer **222**. Therefore, light of the vertical shadow area of the patterned bonding layer **222** is reflected and deviates from this shadow area, thus eliminating light reflection of the patterned bonding layer **222** to the inside parts of the chip. The front recessed array **261** is only a preferred solution, and the present invention is not limited to this method. In other embodiments, it may also be a series of irregularly arranged recessed structures by an etching depth is 20~100 nm, preferably 30~70 nm.

(35) Although specific embodiments have been described above in detail, the description is merely for purposes of illustration. It should be appreciated, therefore, that many aspects described above are not intended as required or essential elements unless explicitly stated otherwise. Various

modifications of, and equivalent acts corresponding to, the disclosed aspects of the exemplary embodiments, in addition to those described above, can be made by a person of ordinary skill in the art, having the benefit of the present disclosure, without departing from the spirit and scope of the disclosure defined in the following claims, the scope of which is to be accorded the broadest interpretation so as to encompass such modifications and equivalent structures.

Claims

1. A flip-chip light emitting diode comprising: a light-emitting epitaxial laminated layer having a first surface, and a second surface opposing the first surface, and including a first semiconductor layer, an active layer, and a second semiconductor layer, wherein the first surface is a roughened surface; a transparent bonding medium layer over the first surface of the light-emitting epitaxial laminated layer and bonded with a transparent substrate; a first electrode and a second electrode respectively over a first electrode region and a second electrode region over the second surface of the light-emitting epitaxial laminated layer; wherein the transparent substrate has a thickness of less than 80 μm ; wherein the transparent bonding medium layer includes a first sub-layer and a second sub-layer, wherein the first sub-layer is adjacent to the first surface of the light-emitting epitaxial laminated layer, and the second sub-layer is adjacent to the transparent substrate; wherein the first sub-layer has a refractive index larger than a refractive index of the second sub-layer; wherein the first sub-layer has a thickness less than 1 μm and comprises at least one of Al_2O_3 or SiN_x , configured to reduce a refractive index difference between the light-emitting epitaxial laminated layer and the transparent bonding medium layer, thereby improving a light extraction efficiency, wherein the second sub-layer has a thickness of more than 1 μm and comprises a silicon oxide layer; wherein the transparent bonding medium layer has a flat upper surface distal from the transparent substrate; and wherein the second sub-layer has a circular array structure.
2. The flip-chip light emitting diode of claim 1, wherein the transparent bonding medium layer has a thickness of 1.5~4 μm .
3. The flip-chip light emitting diode of claim 1, wherein the transparent bonding medium layer has the flat upper surface proximal to the transparent substrate and a patterned lower surface distal from the transparent substrate.
4. The flip-chip light emitting diode of claim 3, wherein the flat upper surface has a roughness value $R_a < 1 \text{ nm}$.
5. The flip-chip light emitting diode of claim 1, wherein the second surface of the light-emitting epitaxial laminated layer has a portion which is etched to form a recess structure.
6. The flip-chip light emitting diode of claim 5, wherein the recess structure has a depth of 20~100 nm.
7. A flip-chip light emitting diode comprising: a light-emitting epitaxial laminated layer having a first surface and a second surface opposing the first surface, and including a first semiconductor layer, an active layer and a second semiconductor layer; a transparent bonding medium layer over the first surface of the light-emitting epitaxial laminated layer and bonded with a transparent substrate; a first electrode and a second electrode respectively over a first electrode region and a second electrode region over the second surface of the light-emitting epitaxial laminated layer; wherein the transparent substrate has a thickness of less than 80 μm ; wherein the transparent bonding medium layer comprises a high-refractivity transparent medium layer and a transparent bonding layer, wherein: the high-refractivity transparent medium layer is over the first surface of the light-emitting epitaxial laminated layer; and the transparent bonding layer is over the high-refractivity transparent medium layer; wherein the high-refractivity transparent medium layer has a thickness less than 1 μm and comprises at least one of Al_2O_3 or SiN_x , configured to reduce a refractive index difference between the light-emitting epitaxial laminated layer and the transparent bonding medium layer, thereby improving a light extraction efficiency; wherein the

transparent bonding layer has a thickness of more than 1 μm and comprises a silicon oxide layer; wherein the transparent bonding layer has a flat upper surface distal from the transparent substrate; and wherein the transparent bonding layer has a circular array structure.

8. A light-emitting device comprising one or more light emitting diodes including at least one flip-chip light emitting diode according to claim 7.

9. The light-emitting device of claim 8, wherein the second surface of the light-emitting epitaxial laminated layer has a portion which is etched to form a recess structure.

10. The light-emitting device of claim 8, wherein the first surface of the light-emitting epitaxial laminated layer is a roughened surface.

11. The flip-chip light emitting diode of claim 7, wherein the high-refractivity transparent medium layer is composed of Al_2O_3 ; and the transparent bonding layer is composed of SiO_2 .

12. The flip-chip light emitting diode of claim 7, wherein the first surface of the light-emitting epitaxial laminated layer is a roughened surface.

13. The flip-chip light emitting diode of claim 12, wherein the transparent bonding layer has the flat surface adjacent proximal to the transparent substrate, and a patterned lower surface distal from the transparent substrate.

14. The flip-chip light emitting diode of claim 7, wherein the second surface of the light-emitting epitaxial laminated layer has a portion which is etched to form a recess structure.

15. The flip-chip light emitting diode of claim 7, wherein the transparent bonding layer has a thickness of 1~2 μm .

16. A light-emitting device comprising one or more light emitting diodes including at least one light emitting diode comprising: a light-emitting epitaxial laminated layer having a first surface and a second surface opposing the first surface, and including a first semiconductor layer, an active layer and a second semiconductor layer, wherein the first surface is a roughened surface; a transparent bonding medium layer over the first surface of the light-emitting epitaxial laminated layer and bonded with a transparent substrate; a first electrode and a second electrode respectively over a first electrode region and a second electrode region over the second surface of the light-emitting epitaxial laminated layer; wherein the transparent substrate has a thickness of less than 80 μm ; wherein the transparent bonding medium layer including a first sub-layer and a second sub-layer, wherein the first sub-layer is adjacent to the first surface of the light-emitting epitaxial laminated layer, and the second sub-layer is adjacent to the transparent substrate; wherein the first sub-layer has a refractive index larger than a refractive index of the second sub-layer; wherein the first sub-layer has a thickness less than 1 μm and comprises at least one of Al_2O_3 or SiN_x , configured to reduce a refractive index difference between the light-emitting epitaxial laminated layer and the transparent bonding medium layer, thereby improving a light extraction efficiency; wherein the second sub-layer has a thickness of more than 1 μm and comprises a silicon oxide layer; wherein the transparent bonding medium layer has a flat upper surface distal from the transparent substrate; and wherein the second sub-layer has a circular array structure.

17. The light-emitting device of claim 16, wherein the first surface of the light-emitting epitaxial laminated layer is a roughened surface.
